APSC 2113 - Engineering Analysis I

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Homework 1 - Fall 2026

Homework Instructions

- The homework is due on **September 17**.
- Answer all questions. Simplify all answers completely.
- You may collaborate and discuss the homework with other students. However, each student must submit their own answers in their own submission.
- You may use the class notes, textbook, and course notes available on the course webpage to complete the homework. The use of AI tools is not permitted for this homework.
- Write your name and student ID clearly at the top of the page.

Questions

Q1. Find a function y=f(x) satisfying the given differential equation and the prescribed initial condition. **[1 pt]**

$$\frac{dy}{dx} = 2x + 1 \quad y(0) = 3$$

Q2. Find a function y=f(x) satisfying the given differential equation and the prescribed initial condition. **[1 pt]**

$$\frac{dy}{dx} = cos2x \quad y(0) = 1$$

Q3. A diesel car gradually speeds up so that for the first 10 s its acceleration is given by

$$\frac{dv}{dt} = (0.12)t^2 + (0.6)t \quad (ft/s^2)$$

If the car starts from rest ($x_0 = 0, v_0 = 0$) where $\frac{dx}{dt} = v$, find the distance it has traveled at the end of the first 10 s and its velocity at that time. **[1.5 pt]**

Q4. Construct the direction plots for appropriate values. Solve the problem analytically for the complete solution. Show that the particular solution is one of the isoclines determined for the direction plot. **[1.5 pt]**

$$y' = x + y; \quad y(0) = 0; \quad y(-4) = ?$$

Q5. Suppose you are in a helicopter hovering at an altitude of 3000 feet. You bail out of the helicopter and pull the ripcord of your parachute. Your downward velocity satisfies the initial value problem. Find the complete solution for this problem. **[1.5 pt]**

$$\frac{dv}{dt} = 10 - 10v \quad v(0) = 0.$$

Q6. Find the general solutions of the given differential equation. **[1 pt]**

$$\frac{dy}{dx} = \frac{(x-1)y^5}{x^2(2y^3 - y)}$$

Q7. Find the general solutions of the given initial value problem. **[1 pt]**

$$\frac{dy}{dx} = ye^x \quad y(0) = 2e$$

Q8. Find the general solution of the given initial value problem. **[1 pt]**

$$2y\frac{dy}{dx} = \frac{x}{\sqrt{x^2 - 16}} \quad y(5) = 2$$

Q9. Upon the birth of their first child, a couple deposited $5000 in an account that pays 8% interest compounded continuously. The interest payments are allowed to accumulate. How much will the account contain on the child's eighteenth birthday? **[1.5 pt]**

Q10. Suppose that you discover in your attic an overdue library book on which your grandfather owed a fine of 30 cents 100 years ago. If an overdue fine grows exponentially at a 5% annual rate compounded continuously, how much would you have to pay if you returned the book today? **[1.5 pt]**

Q11. Suppose that sodium pentobarbital is used to anesthetize a dog. The dog is anesthetized when its bloodstream contains at least 45 milligrams (mg) of sodium pentobarbitol per kilogram of the dog's body weight. Suppose also that sodium pentobarbitol is eliminated exponentially from the dog's bloodstream, with a half-life of 5 h. What single dose should be administered in order to anesthetize a 50-kg dog for 1 h? **[1.5 pt]**
Note: In many cases the amount A(t) of a certain drug in the bloodstream, measured by the excess over the natural level of the drug, will decline at a rate proportional to the current excess amount. That is,

$$\frac{dA}{dt} = -\lambda A$$

where $\lambda > 0$. The parameter λ is called the elimination constant of the drug.

Q12. The barometric pressure p (in inches of mercury) at an altitude x miles above sea level satisfies the initial value problem

$$\frac{dp}{dx} = (-0.2)p \quad p(0) = 29.92.$$

(a) Calculate the barometric pressure at 10,000 ft and again at 30,000 ft.
(b) Without prior conditioning, few people can survive when the pressure drops to less than 15 in. of mercury. How high is that? **[1.5 pt]**

Q13. A cake is removed from an oven at 210° F and left to cool at room temperature, which is 70° F. After 30 min the temperature of the cake is 140° F. When will it be 100° F? **[1.5 pt]**
Note: According to Newton's law of cooling, the time rate of change of the temperature T(t) of a body immersed in a medium of constant temperature A is proportional to the difference A - T . That is,

$$\frac{dT}{dt} = k(A - T)$$

where k is a positive constant.

Q14. Find the general solution of the given differential equation **[1 pt]**

$$x\frac{dy}{dx} = 2y$$

Q15. Find the complete solution of the given differential equation **[1 pt]**

$$\frac{dy}{dx} = -2y + e^{-x}$$

Q16. Find the particular solution of the given initial value problem **[1 pt]**

$$\frac{dy}{dx} = 3x + e^{x} \quad y(0) = 1 + \pi$$

(1) $\frac{dy}{dx} = 2x + 1 \qquad y(0) = 3$

$$\int dy = \int (2x+1)\, dx + C$$

$$y = x^2 + x + C$$

At $y(0) = 3 \Rightarrow C = 3$

$$\boxed{y(x) = x^2 + x + 3}$$

(2) $\frac{dy}{dx} = \cos 2x \qquad y(0) = 1$

$$\int dy = \int \cos 2x\, dx + C$$

$$y = \frac{\sin 2x}{2} + C$$

$y(0) = 1 \Rightarrow C = 1$

$$\boxed{y(x) = \frac{\sin 2x}{2} + 1}$$

(3) $\frac{dv}{dt} = (0.12)t^2 + (0.6)t$

$$\int dv = \int 0.12t^2 + 0.6t\, dt + C_1$$

$$v(t) = 0.04t^3 + 0.3t^2 + C_1$$

At $t = 0, v_0 = 0 \Rightarrow C_1 = 0$

$$\boxed{\begin{aligned} v(t) &= 0.04t^3 + 0.3t^2 \\ v(10) &= 70 \text{ ft/s} \end{aligned}}$$

$$v(t) = 0.04t^3 + 0.3t^2$$

$$\frac{dx}{dt} = 0.04t^3 + 0.3t^2$$

$$\int dx = \int (0.04t^3 + 0.3t^2)\, dt + c_2$$

$$x(t) = 0.01t^4 + 0.1t^3 + c_2$$

At $t = 0$, $x_0 = 0$ $\Rightarrow$ $c_2 = 0$

$$x(t) = 0.01t^4 + 0.1t^3$$

$$x(10) = 200 \text{ ft}$$

(4) $y' = y + x$ $\quad y(0) = 0$ $\quad y(-4) = ?$

ISOCLINE EQUATION

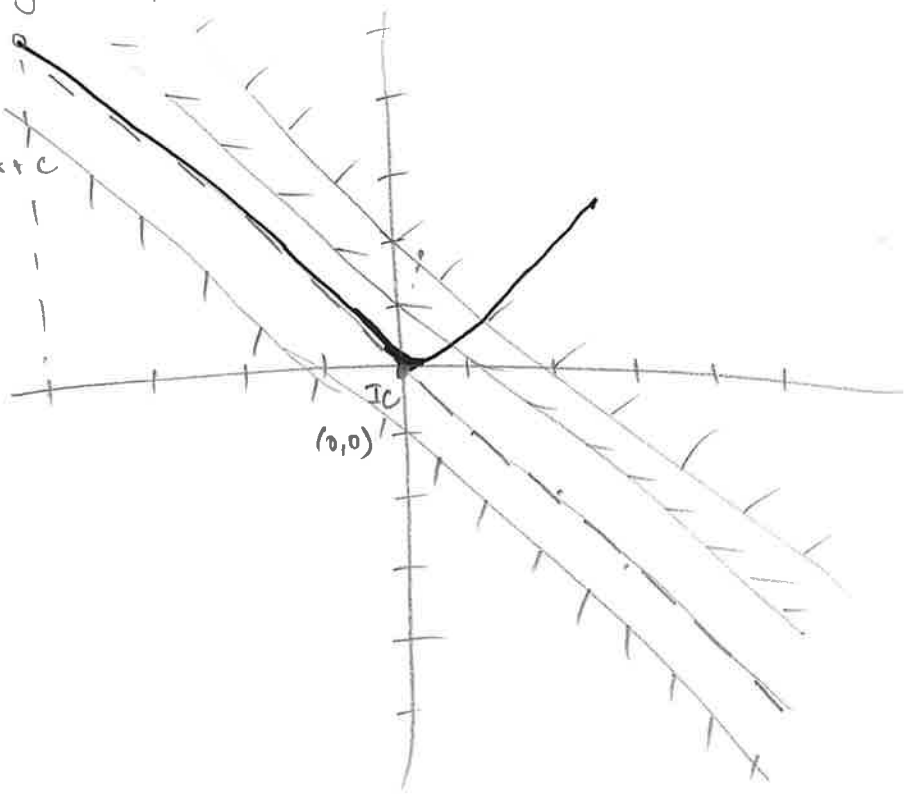
$x + y = c \Rightarrow y = -x + c$

$c = -1$ $\quad y = -x - 1$

$c = 0$ $\quad y = -x$

$c = 1$ $\quad y = -x + 1$

$c = 2$ $\quad y = -x + 2$



$y' = y + x$

Particular Soln:

Assume $y_p = Ax + B$

$$\frac{dy_p}{dx} = y_p + x$$

$$\frac{d(Ax+B)}{dx} = (Ax + B) + x$$

$$A + 0 = Ax + x + B$$

$$A - B = Ax + x = (A+1)x$$

$$0 = (A+1)x + (B - A)$$

Solving for constants,

$$A + 1 = 0 \Rightarrow A = -1$$

$$B - A = 0 \Rightarrow B = A = -1$$

$$\boxed{y_p = -x - 1}$$

This matches the isocline for $C = -1$

Homogeneous soln:

$$\frac{dy_h}{dx} = y_h$$

$$\int \frac{dy_h}{y_h} = \int dx + C$$

$$\log y_h = x + C$$

$$y_h = ce^x$$

Complete Solution:

$$y(x) = (-x-1) + ce^{x}$$

Initial Condition: $y(0) = 0$

$$0 = (-0-1) + ce^{0} \quad \Rightarrow \quad c = 1$$

$$\boxed{y(x) = -1 - x + e^{x}}$$

$$y(-4) = e^{-4} + 3$$
$$= 3.0183$$

(5)

$$\frac{dv}{dt} = 10 - 10v \qquad v(0) = 0$$

Particular Soln

Assume $y_p = A$

$$\frac{dy_p}{dt} = -10y_p + 10$$

$$\frac{dA}{dt} = -10A + 10$$

$$0 = -10A + 10$$

$$A = \frac{-10}{-10} = 1$$

$$y_p = 1$$

Homogenous Soln

$$\frac{dy_h}{dt} = -10\,y_h$$

$$\int \frac{dy_h}{y_h} = \int -10\,dt + C$$

$$\ln y_h = -10t + C$$

$$y_h = ce^{-10t}$$

Complete Soln

$$y = 1 + ce^{-10t}$$

At $x=0, y=0$

$$0 = 1 + c. \Rightarrow c = -1$$

$$\boxed{y = 1 - 1e^{-10t}}$$

(6) $$\frac{dy}{dx} = \frac{(x-1)y^5}{x^2(2y^3 - y)}$$

$$\frac{2y^3 - y}{y^5}\,dy = \frac{x-1}{x^2}\,dx$$

$$\int (2y^{-2} - y^{-4})\,dy = \int (x^{-1} - x^{-2})\,dx + C$$

$$\frac{2y^{-1}}{-1} - \frac{y^{-3}}{-3} = \ln x - \frac{x^{-1}}{-1} + C$$

$$\boxed{\frac{-2}{y} + \frac{1}{3y^3} = \ln x + \frac{1}{x} + C}$$

(7) $\frac{dy}{dx} = ye^x$ $\quad y(0) = 2e$

$$\int \frac{dy}{y} = \int e^x dx + C$$

$$\log y = e^x + C$$

$$y = ce^{e^x}$$

For $y(0) = 2e$

$$2e = ce^{e^0} \Rightarrow \quad c = 2$$

$$\boxed{y = 2e^{e^x}}$$

(8) $2y \frac{dy}{dx} = \frac{x}{\sqrt{x^2-16}}$ $\quad y(5) = 2$

$$\int 2y\,dy = \int \frac{x}{\sqrt{x^2-16}}\,dx + C$$

$$y^2 = \int \frac{x}{\sqrt{a}} \cdot \frac{da}{2x} + C$$

$$= \frac{1}{2} \cdot 2\sqrt{x^2-16} + C$$

$$y^2 = \sqrt{x^2-16} + C$$

For $y(5) = 2$

$$(2)^2 = \sqrt{5^2-16} + C$$

$$c = 1$$

$$\boxed{y^2 = \sqrt{x^2-16} + 1}$$

$a = x^2 - 16$

$\frac{da}{dx} = 2x \Rightarrow dx = \frac{da}{2x}$

$\int \frac{da}{\sqrt{a}} = 2\sqrt{a}$

(9) Given

$a = 8\% = 0.08$

$q(t) = 0$

IC: $y(0) = 5000$ \$

$$\frac{dy}{dt} = ay + q(t)^{>0}$$

$$\frac{dy}{dt} = 0.08\,y$$

$$\int \frac{dy}{y} = \int 0.08\,dt + c$$

$$y(t) = ce^{0.08t}$$

For $y(0) = 5000$, $c = 5000$

$$y(t) = 5000\,e^{0.08t}$$

At $t = 18$,

$$y(18) = 5000\,e^{0.08(18)}$$

$$y(18) = \$21,103.5$$

(10) Given

$a = 5\% = 0.05$

$y(0) = 0.3$ \$

$$\frac{dy}{dt} = ay + q(t)^{>0}$$

$$\frac{dy}{y} = a\,dt$$

$$\int \frac{dy}{y} = \int a\,dt + C$$

$$\ln y = at + C$$

$$y(t) = Ce^{at}$$

For $y(0) = 0.3$, $C = 0.3$

$$y(t) = 0.3e^{at}$$

$$y(100) = 44.52 \quad \#$$

(11) $A(0)$ For 1 kg dog $\cong$ 45 mg

$A(0)$ for 50 kg dog = 2250 mg.

$$\frac{dA}{dt} = -\lambda A$$

$$\int \frac{dA}{A} = \int -\lambda\,dt + C$$

$$\ln A = -\lambda t + C$$

$$A(t) = Ce^{-\lambda t}$$

Since $A(0)$ for 50 kg $= 2250 \Rightarrow C = 2250$

$$A(t) = 2250\, e^{-\lambda t}$$

Given Half-life = 5 hours

$\Rightarrow t = 5$ hrs for $\frac{A(0)}{A(5)} = \frac{1}{2}$

$$\frac{1}{2} = \frac{2250\, e^{-\lambda(0)}}{2250\, e^{-\lambda(5)}}$$

$$\frac{1}{2} = e^{-5\lambda}$$

$$-5\lambda = \ln(0.5)$$

$$\lambda = 0.1386 \ h^{-1}$$

For 1 hr, $A(1) = 2250e^{-0.1386(1)} = 418.5$

Dosage should be minimum dose + 418.5

$= 2250 + 418.5$

$= 2584.5$ mg

(12) $\frac{dp}{dx} = -0.2p \qquad p(0) = 29.92$

$$\int \frac{dp}{p} = \int -0.2\, dx + C$$

$$\log p = Ce^{-0.2x}$$

For $p(0) = 29.92$

$$29.92 = ce^{0}$$

$$c = 29.92$$

$$p = 29.92\, e^{-0.2x}$$

(a) At 10000 ft,

$x = 1.89$ mi

$$p = 29.92\, e^{-0.2 \times 1.89}$$

$= 20.5$ in Hg

At 30000 ft,

$x = 5.68$ mi

$$p = 29.92\, e^{-0.2 \times 5.68}$$

$= 9.61$ in Hg

(b) $p = 15$ in

$$e^{-0.2x} = \frac{15}{29.92} = 0.501$$

$$-0.2x = \ln(0.501) = -0.691$$

$x = 3.45$ mi

$= 18216$ ft

(13) Given Room temp A = 70 F

$T(0) = 210$ F

$T(30) = 140$ F

$T(x) = 100$ F

$$\frac{dT}{dt} = k(A - T)$$

$$\int \frac{dT}{A - T} = \int k\,dt + C$$

$$-1 \ln(A - T) = kt + C$$

$$A - T = Ce^{-kt}$$

$$T = A - Ce^{-kt}$$

At $T(0) = 210$ F,

$$210 = 70 - Ce^{-k(0)}$$

$$C = -140$$

$$T(t) = 70 + 140e^{-kt}$$

At $T(30) = 140$ F

$$140 = 70e^{-30k}$$

$$k = 0.0231$$

$$T(t) = 70 + 140e^{-0.0231t}$$

At $T(x) = 100$

$$100 = 70 + 140e^{-0.0231t}$$

$$t = 66.74 \text{ min}$$

$$= 66 \text{ min } 44 \text{ sec}$$

(14)

$$x \frac{dy}{dx} = 2y$$

Particular Soln

Assume $y_p = A$

$$x \frac{dy_p}{dx} = 2y_p$$

$$x \frac{dA}{dx}^{0} = 2A$$

$$A = 0$$

$$y_p = 0$$

Homogenous Soln

$$x \frac{dy_h}{dx} = 2y_h$$

$$\int \frac{dy_h}{y_h} = \int \frac{2}{x} dx + C$$

$$\ln y_h = 2 \ln x + C$$

$$= \ln x^2 + C$$

$$y_h = Cx^2$$

Complete Soln

$$y = y_p + y_h$$

$$y = Cx^2$$

1) $\frac{dy}{dx} = -2y + e^{-x}$

Particular Soln

Assume $y_p = Ye^{-x}$

$$\frac{dy_p}{dx} = -2y_p + e^{-x}$$

$$-Ye^{-x} = -2Ye^{-x} + e^{-x}$$

$$-Y = -2Y + 1$$

$$Y = 1$$

$$y_p = e^{-x}$$

Homogenous Soln

$$\frac{dy_h}{dx} = -2y_h$$

$$\int \frac{dy_h}{y_h} = \int -2dx + C$$

$$\ln y_h = -2x + C$$

$$y_h = Ce^{-2x}$$

Complete Soln

$$y = y_p + y_h$$

$$y = e^{-x} + Ce^{-2x}$$

(16)

$$\frac{dy}{dx} = 3x + e^x \qquad y(0) = 1 + \pi$$

(15)

Particular Soln

$$\int dy_p = \int (3x + e^x)\,dx + C$$

$$y_p = \frac{3x^2}{2} + e^x + C$$

$$y_p(0) = 1 + C$$

$$1 + \pi = 1 + C$$

$$C = \pi$$

$$\boxed{y_p = \frac{3x^2}{2} + e^x + \pi}$$